

Literature Review

Streaming AIOps, Self-Healing Data Pipelines
and Agentic Frameworks on Commodity Hardware

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Papers Reviewed

[Survey] Mayilsamy, M. (2025). *A Comprehensive Survey of Streaming Large Language Models: Architectures, Applications, and Future Directions*. Journal of Information Systems Engineering and Management, 10(60s).

[Methodology] Kirubakaran, A. M. et al. (2025). *Governing Cloud Data Pipelines with Agentic AI*. arXiv preprint arXiv:2512.23737.

[Methodology] Chakraborty, S. (2025). *Beyond ETL: How AI Agents Are Building Self-Healing Data Pipelines*. Journal of Computer Science and Technology Studies, 7(3), 741–756.

[Conceptual] Kothamasu, L. S. (2025). *Autonomous Resilience: Advancing Data Engineering Through Self-Healing Pipelines and Generative AI*. European Journal of Computer Science and Information Technology, 13(28), 102–113.

1. Overview

This literature review surveys four papers at the intersection of streaming data pipeline engineering, self-healing systems, agentic AI, and LLM deployment on constrained hardware. Each paper is analysed across three dimensions: the research gap it identifies, the methodology it proposes, and the future directions it admits. A final synthesis section positions our own final year project contribution directly against these findings, demonstrating where each paper stops and where our work begins.

The review is structured around the task requirement: one survey paper and three supporting works (two methodology-oriented, one conceptual). Taken together, these four papers define the state of the art as of early 2025 and expose a consistent and critical blind spot — the absence of any implemented, measured, end-to-end agentic self-healing system running on real distributed hardware without cloud or GPU dependency.

2. Individual Paper Analyses

Paper 1 · Survey

A Comprehensive Survey of Streaming Large Language Models: Architectures, Applications, and Future Directions

Mayilsamy, M. · JISEM, 10(60s), 2025 · Survey

Research Gap Identified	Methodology	Future Work Proposed
Most enterprise AI deployments still rely on batch processing pipelines. Existing literature on streaming systems treats LLMs as offline components, not as real-time inference engines embedded within live data flows. No unified survey existed that mapped the transition from batch to stream for LLMs, catalogued the latency-throughput trade-offs specific to continuous inference, or synthesised enabling technologies (quantisation, distillation, edge-cloud hybrid deployment) under one framework.	Systematic literature review across IEEE Xplore, ACM DL, arXiv, and Springer (Jan 2020 – Aug 2025). Initial corpus of 183 papers narrowed to 67 core works through multi-stage screening. Classification taxonomy of four axes: Architectural Approach, Optimisation Targets, Application Domain, and Deployment Context. Synthesises inference optimisation techniques (continuous batching, KV cache management, speculative decoding, PagedAttention), stream processing frameworks (Kafka, Flink, Pulsar), and enabling hardware/software techniques (distillation, 4-bit quantisation, edge-cloud split).	Adaptive/continual learning without catastrophic forgetting in streaming contexts. Multi-modal integration (text + audio + sensor) for richer real-time understanding. Explainability mechanisms for streaming inference decisions. Benchmarks that capture long-duration stream performance, not just single-request latency. Hardware-aware optimisation strategies for resource-constrained edge deployments.

Relevance to Our Research

The survey directly validates two core architectural choices in our system. First, it confirms that 4-bit quantisation (used for Phi-4 Mini on Node 2) is a production-viable technique for resource-constrained deployments, achieving memory reductions of approximately 70% with acceptable accuracy degradation. Second, the RabbitMQ-based asynchronous decoupling in our architecture mirrors the survey's central recommendation: never embed heavy LLM inference directly into the synchronous data path. Our commodity-hardware benchmark results will also contribute to the gap the survey identifies — the absence of long-duration, real-hardware streaming LLM performance data.

Paper 2 · Methodology

Governing Cloud Data Pipelines with Agentic AI

Kirubakaran, A. M. et al. · arXiv:2512.23737 · 2025 · Methodology

Research Gap Identified	Methodology	Future Work Proposed
Cloud data pipelines rely on static orchestration frameworks (Airflow, Prefect, Dagster) that manage when tasks run but not how pipelines should adapt dynamically. Failures from schema drift, upstream instability, resource contention, and workload spikes require manual diagnosis — inflating MTTR, cloud costs, and reducing data freshness. Rule-based automation systems are brittle at scale. Existing LLM-driven autonomous agents introduce safety and compliance risks because they operate without governance constraints.	Proposes Agentic Cloud Data Engineering (ACDE): a policy-aware control-plane architecture with three planes — Data Plane (execution), Agentic Control Plane (four specialised agents: Monitoring, Optimisation, Schema, Recovery), and Policy & Governance Plane (declarative rules enforced before any action). Agents operate in an observe-reason-propose-evaluate loop; they propose actions but cannot execute without policy validation. Evaluated using TPC-DS batch pipelines and NYC Taxi streaming workloads in a managed cloud environment. LLM backends (GPT, Claude, Gemini) used interchangeably as bounded reasoning engines — policy layer provides safety guarantees regardless of model.	Explicit multi-agent coordination for pipelines with complex cross-dependencies (current agents are loosely coupled). Policy learning from historical execution data to reduce manual governance configuration. Extension to hybrid and multi-cloud environments with heterogeneous observability. Formal verification of policy-agent interaction guarantees. No on-premise or commodity hardware evaluation is discussed.

Relevance to Our Research

This paper is the closest architectural relative to our system and is our primary contrast point. ACDE demonstrates a 45% MTTR reduction, 25% cost reduction, and 70% fewer manual interventions — but exclusively in a cloud environment using managed compute clusters and GPU-backed frontier LLMs (GPT, Claude, Gemini). It explicitly notes that commodity hardware deployment is out of scope. Our contribution is to fill exactly that gap: we implement the same governance-bounded agentic control paradigm on three CPU-only nodes using a 4-bit quantised Phi-4Mini, producing the first empirical benchmark of whether policy-aware multi-agent control remains effective under severe resource constraints. Additionally, ACDE explicitly identifies multi-agent coordination as future work — our four-agent architecture with explicit RabbitMQ handoffs and per-agent latency measurement is a direct answer to that open problem.

Paper 3 · Methodology

Beyond ETL: How AI Agents Are Building Self-Healing Data Pipelines

Chakraborty, S. · JCSTS, 7(3), 741–756 · 2025 · Methodology

Research Gap Identified	Methodology	Future Work Proposed
Traditional ETL pipelines rely on static rules, reactive maintenance cycles, and human-in-the-loop troubleshooting that cannot scale with modern data volumes or velocity. Organisations spend 40–60% of engineering time on maintenance rather than innovation. Silent failures (structural and contextual) propagate through analytics ecosystems before detection. Existing literature documents these operational costs but treats AI intelligence as a monolithic entity — no work decomposes the intelligence into specialised agents with distinct roles, nor addresses the computational latency created by placing heavy AI reasoning on the synchronous data path.	Conceptual-empirical hybrid. Synthesises industry case studies and secondary research to document the performance of horizontal agents (cross-pipeline oversight: anomaly detection, resource optimisation, dependency mapping) and vertical agents (domain-specific expertise: schema validation, compliance enforcement, metadata reconciliation, data quality). Presents quantified performance claims from cited industry sources: 87% early detection rate, 65% auto-resolution of data quality issues, 76% reduction in pipeline failures, 83% MTTR reduction, 94% detection accuracy via model retraining. Does not describe an implemented prototype or controlled experiment — all figures are drawn from existing third-party deployments.	Cognitive pipelines that understand business context to make semantically-aware prioritisation decisions. Natural language interfaces to democratise pipeline management for non-technical users. Cross-organisational federated learning for collective anomaly resolution knowledge. Predictive infrastructure scaling combining time-series forecasting with strategic planning signals. No implementation roadmap or benchmark framework is proposed.

Relevance to Our Research

This paper provides the strongest conceptual justification for our horizontal/vertical agent decomposition. Chakraborty's distinction between cross-pipeline oversight (analogous to our Policy Agent's cluster-wide dispatch decisions) and domain-specific agents (analogous to our Triage and Strategy agents) directly maps to our four-agent architecture. However, the paper's complete absence of a working implementation and controlled evaluation is precisely the gap we fill. Our 1,950-event synthetic dataset, Cohen's Kappa strategy evaluation, and per-agent RabbitMQ timestamp benchmarks produce the empirical results that Chakraborty's framework requires but does not provide. The Learning Agent in our Layer 2 also operationalises Chakraborty's 'knowledge accumulation' and 'policy refinement' concepts, which remain purely theoretical in this paper.

Paper 4 · Conceptual

Autonomous Resilience: Advancing Data Engineering Through Self-Healing Pipelines and Generative AI

Kothamasu, L. S. · EJCSIT, 13(28), 102–113 · 2025 · Conceptual

Research Gap Identified	Methodology	Future Work Proposed
Traditional data pipelines are reactive, statically configured, and require extensive manual intervention for error resolution. They lack predictive capabilities and cannot adapt to schema evolution without reconfiguration. Self-healing mechanisms represent an advancement, but existing treatments rely entirely on Generative AI across all pipeline layers — including the synchronous data path — without acknowledging the latency penalty this introduces. No framework separates fast statistical detection from heavy generative reasoning.	Conceptual framework paper. Proposes a layered architecture for self-healing pipelines: monitoring, diagnostic, decision-making, and execution components forming a feedback loop. Surveys applicable generative AI models (transformers, VAEs, GANs, reinforcement learning) and their pipeline applications. Discusses implementation strategies, scalability challenges, governance considerations, and organisational transformation. Provides no experimental evaluation, no prototype system, no benchmark dataset, and no performance measurements. All quantitative claims reference secondary or tertiary industry sources rather than original experiments.	Progressing from reactive anomaly detection to comprehensive predictive and adaptive capabilities. Ethical oversight frameworks for autonomous data systems. Integration with legacy data ecosystems through phased deployment. Federated governance and data mesh approaches. The paper explicitly acknowledges that performance limitations of Generative AI in low-latency streaming environments remain an open engineering challenge — but does not propose a resolution.

Relevance to Our Research

Kothamasu provides the broadest conceptual framing for why self-healing pipelines matter and what generative AI contributes to them. However, it is also the weakest in empirical grounding — the paper explicitly acknowledges that relying entirely on Generative AI introduces performance limitations in streaming environments, yet proposes no solution to this problem. This is the precise tension our architecture resolves through decoupling: Isolation Forest and Pydantic-based anomaly detection on the synchronous data path (Layer 1) never waits for LLM inference, while Phi-4 Mini reasoning operates asynchronously in Layer 2. Our system is the implementation of the solution Kothamasu identifies as necessary but does not build.

3. Cross-Paper Gap Analysis

Reading the four papers together reveals not just individual gaps but a systemic pattern in the 2025 state of the art. The following table maps each paper's contribution against five dimensions that define a complete, deployable, and evaluable AIOps system.

Dimension	Mayilsamy (Survey)	Kirubakaran (ACDE)	Chakraborty (ETL)	Kothamasu (AR)	Our Project
Working Implementation	No (survey)	Partial (cloud managed)	No (conceptual)	No (conceptual)	Yes — 3-node cluster
Commodity / CPU-only Hardware	Discussed	No — cloud GPU	Not addressed	Not addressed	Yes — AMD Ryzen + i5
Multi-Agent Decomposition	Not in scope	4 agents (no explicit coordination)	H/V agent concepts	Not specified	4 agents + RabbitMQ coordination
Empirical Benchmark Results	Secondary sources only	Yes — cloud environment	No — industry anecdotes	No — secondary sources	Yes — real hardware
Human-in-the-Loop Gateway	Not in scope	Operator approval mentioned	Mentioned conceptually	Governance discussed	Fully implemented (Django)
Adaptive Learning / Feedback Loop	Future direction	Policy learning — future work	Knowledge accumulation — conceptual	Continuous improvement — conceptual	Learning Agent — implemented
Quantised Edge LLM Evaluation	Discussed in theory	No — frontier LLMs only	Not addressed	Not addressed	Phi-4 Mini 4-bit on CPU

The Unified Gap Statement — for use in Section 1 of our paper

Mayilsamy (2025) surveys the transition to streaming LLM architectures and identifies quantisation and edge deployment as viable directions but provides no implementation evidence. Chakraborty (2025) documents the operational benefits of horizontal and vertical AI agents in self-healing pipelines and defines the conceptual architecture for multi-agent decomposition, but treats AI as a monolithic entity and presents no working system or benchmark results. Kothamasu (2025) proposes a layered generative AI architecture for self-healing pipelines but explicitly acknowledges unresolved performance limitations in low-latency streaming environments and contributes no experimental data. Kirubakaran et al. (2025) is the most technically rigorous of the four works, demonstrating a 45% MTTR reduction and 70% reduction in manual interventions through policy-bounded agentic control — but exclusively in a cloud environment with managed compute and GPU-backed frontier LLMs.

No existing work implements, deploys, or benchmarks a multi-agent self-healing pipeline on real distributed commodity hardware. This project directly addresses that gap.

4. Methodology Synthesis

Across the three implementation-oriented papers, four methodological patterns emerge that directly inform design decisions in our system. Each is examined below.

4.1 Decoupled Synchronous / Asynchronous Processing

Both Kirubakaran et al. and the Mayilsamy survey converge on the same architectural principle: LLM inference must never block the synchronous data path. Kirubakaran implements this through a separate Agentic Control Plane that consumes telemetry asynchronously. Mayilsamy documents this as the enabling insight behind production streaming LLM deployments. Chakraborty and Kothamasu identify the problem (LLM latency on the data path) but do not resolve it architecturally.

Our Layer 1 and Layer 2 separation — where Layer 1 performs millisecond anomaly detection with Isolation Forest and Pydantic validation while Layer 2 agents reason over RabbitMQ queues asynchronously — is the direct operationalisation of this principle on commodity hardware.

4.2 Policy-Bounded Autonomy

Kirubakaran et al. make the strongest argument for governance-bounded agents rather than unconstrained autonomy. Their Policy Plane validates all proposed actions before execution, ensuring that LLM reasoning cannot bypass compliance rules regardless of which model backend is used. This design insight — separating reasoning capability from execution authority — is essential for enterprise deployment.

Our risk-tier classification in the Policy Agent (low-risk auto-execute vs. high-risk HITL escalation) applies the same principle at a finer granularity. The Policy Agent does not execute actions; it routes them. The Auto-Execution Engine and the HITL Django Gateway hold execution authority. This mirrors Kirubakaran's architecture while making the boundary explicit and measurable through false escalation and false automation rate metrics.

4.3 Evaluation Against Baseline

Kirubakaran et al. is the only paper that includes a proper baseline comparison — static orchestration with fixed resource allocation and human-initiated remediation. Their evaluation demonstrates that agentic control reduces MTTR by 45% and manual interventions by 70% against this baseline.

Our evaluation replicates this baseline design: a threshold-only system (no AI, no agents, static if/else rules) evaluated against the same 1,950-event synthetic dataset. This produces the before/after comparison that is absent from Chakraborty and Kothamasu entirely, and extends Kirubakaran's cloud evaluation to the commodity hardware context.

4.4 Knowledge Accumulation as a System Property

Chakraborty describes knowledge accumulation — the pipeline's ability to learn from past incidents and apply that knowledge to novel failures — as a defining feature of mature self-healing systems. Incident frequency reduction of approximately 32% year-over-year is attributed to systems with effective learning mechanisms. Kothamasu similarly positions continuous learning as the distinguishing feature of generative AI approaches. However, neither paper implements or specifies a concrete learning mechanism.

Our Learning Agent translates this into a concrete, implementable component: per-incident ChromaDB upserts tagged with success/failure outcomes, exponential moving average confidence threshold recalibration, and negative example tagging for rejected HITL decisions. The Learning Agent fires after dispatch — adding zero latency to the remediation loop — and its effectiveness is measured through a windowed Triage recall comparison (early vs. late evaluation windows) over the 1,950-event run.

5. Future Work Synthesis and Research Positioning

The future directions proposed across all four papers collectively define the research frontier. The table below maps each proposed direction to our project's position — either addressing it, partially addressing it, or noting it as out of scope.

Future Direction from Literature	Source	Our Project's Position
Adaptive / continual learning without catastrophic forgetting	Mayilsamy	Partially addressed. Our Learning Agent implements per-incident ChromaDB updates and EMA threshold recalibration — a lightweight form of continual learning without full model fine-tuning.
Multi-agent coordination for complex cross-dependencies	Kirubakaran	Directly addressed. Explicit Triage → Strategy → Policy → Learning coordination over RabbitMQ with per-agent latency measurement and timeout-to-HITL failsafe.
Policy learning from historical execution data	Kirubakaran	Partially addressed. The Learning Agent recalibrates confidence thresholds based on running success rates. Full policy learning (recommendation of new policy rules) is noted as Future Work.
Commodity / edge hardware deployment of agentic systems	All four papers	Core contribution. Three-node AMD Ryzen + Intel i5 cluster, CPU-only, Phi-4 Mini at 4-bit quantisation. First empirical evaluation of this class of system without cloud or GPU dependency.
Benchmarking quantised edge LLMs vs. frontier cloud LLMs	Mayilsamy, Kirubakaran	Optional evaluation dimension. Phi-4 Mini vs. GPT-4o-mini on 30 curated incident prompts using the same Cohen's Kappa rubric. Activated if API access is obtained; noted as Future Work otherwise.
Resolving LLM latency on the synchronous data path	Kothamasu, Chakraborty	Directly addressed. Decoupled synchronous Layer 1 (ML-only detection, sub-10s MTTA target) and asynchronous Layer 2 (LLM reasoning, 30s timeout-to-HITL). The pipeline never waits for LLM inference.
Human-in-the-Loop governance in production systems	All four papers	Directly addressed. Django-based HITL Gateway with full reasoning chain visibility, operator approve/reject, and negative feedback propagation to the Learning Agent.
Multi-cloud and hybrid deployment	Kirubakaran	Out of scope. Our contribution is commodity on-premise. Multi-cloud extension is noted as Future Work.
Formal verification of policy-agent interactions	Kirubakaran	Out of scope. Our Policy Agent uses empirical risk-tier classification. Formal verification is a long-term research direction beyond FYP scope.
Natural language interfaces for pipeline management	Chakraborty, Mayilsamy	Out of scope. Our HITL Dashboard is operator-facing, not NL-driven. NL interface is noted as Future Work.

6. Connection to Our Research Project

6.1 What the Literature Establishes

The four papers collectively establish three foundational facts that our project builds on:

- Agentic AIOps works. Kirubakaran et al. demonstrate that policy-bounded multi-agent control produces measurable improvements in MTTR, cost, and manual intervention frequency when evaluated rigorously. This removes any burden of proof that agents are useful — our paper can take that as established and focus on whether they remain useful under commodity hardware constraints.
- Implementation is the gap. Despite broad agreement on the value of self-healing agents, three of the four papers present no working system and no benchmark results. Chakraborty and Kothamasu explicitly note this limitation. Even Kirubakaran's implementation is cloud-managed and GPU-dependent.
- Quantised edge LLMs are theoretically viable but unmeasured. Mayilsamy documents the enabling technologies (4-bit quantisation, distillation, edge-cloud hybrid) in theory. No paper benchmarks a quantised sub-4B parameter LLM in an AIOps reasoning task on real hardware.

6.2 Our Research Contribution in One Sentence

Primary Contribution

We implement, deploy, and empirically benchmark a four-agent self-healing AIOps system on a three-node commodity CPU-only cluster — producing the first measured evidence of whether policy-bounded multi-agent coordination, human-in-the-loop governance, and adaptive learning remain effective when frontier cloud LLMs and managed infrastructure are replaced by a quantised Phi-4 Mini and off-the-shelf Ethernet-connected hardware.

6.3 Four Independently Citable Contributions

Derived from what the literature leaves open:

- Contribution 1 — Architecture. A fully implemented three-layer tiered hybrid agentic framework for self-healing streaming data pipelines on commodity hardware; the first paper to deploy and benchmark this class of system without cloud or GPU dependency. Fills the implementation gap shared by Chakraborty, Kothamasu, and implicitly identified by Mayilsamy.
- Contribution 2 — Multi-Agent Coordination Protocol. A four-agent system (Triage, Strategy, Policy, Learning) with explicit input/output contracts, event-driven RabbitMQ coordination, per-agent latency measurement, and empirical comparison against a single-agent baseline. Fills the multi-agent coordination gap identified as future work by Kirubakaran.
- Contribution 3 — Adaptive Learning in Production. A Learning Agent that converts ChromaDB from a static RAG store into an evolving knowledge base through per-incident feedback ingestion, operationalising the knowledge accumulation concept described but not implemented by Chakraborty and Kothamasu.
- Contribution 4 — Distributed Commodity Hardware Benchmark. Cross-node latency, RabbitMQ throughput, NTP drift, and end-to-end pipeline timing benchmarks on a real three-node Ethernet-connected cluster. The first empirical dataset of this type for agentic AIOps systems, extending Kirubakaran's cloud evaluation to the edge/commodity context.
- Contribution 5 (Optional) — Edge vs. Cloud LLM Reasoning Quality. A direct empirical comparison of Phi-4 Mini 4-bit vs. GPT-4o-mini on 30 curated AIOps incident prompts, producing the first measured

answer to how close a commodity edge LLM can get to a frontier model on structured reasoning tasks. Activated if API access is obtained.

7. References

- [1] M. Mayilsamy, "A Comprehensive Survey of Streaming Large Language Models: Architectures, Applications, and Future Directions," *Journal of Information Systems Engineering and Management*, vol. 10, no. 60s, 2025.
- [2] A. M. Kirubakaran et al., "Governing Cloud Data Pipelines with Agentic AI," *arXiv preprint arXiv:2512.23737*, Dec. 2025.
- [3] S. Chakraborty, "Beyond ETL: How AI Agents Are Building Self-Healing Data Pipelines," *Journal of Computer Science and Technology Studies*, vol. 7, no. 3, pp. 741–756, May 2025.
- [4] L. S. Kothamasu, "Autonomous Resilience: Advancing Data Engineering Through Self-Healing Pipelines and Generative AI," *European Journal of Computer Science and Information Technology*, vol. 13, no. 28, pp. 102–113, May 2025.